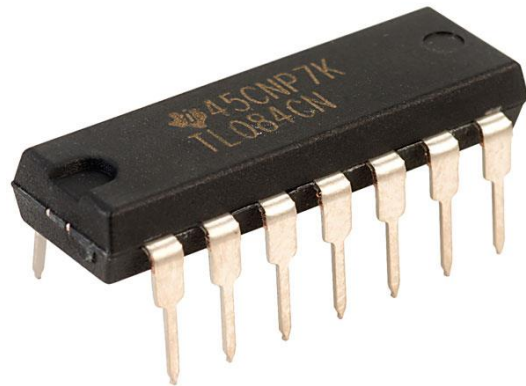
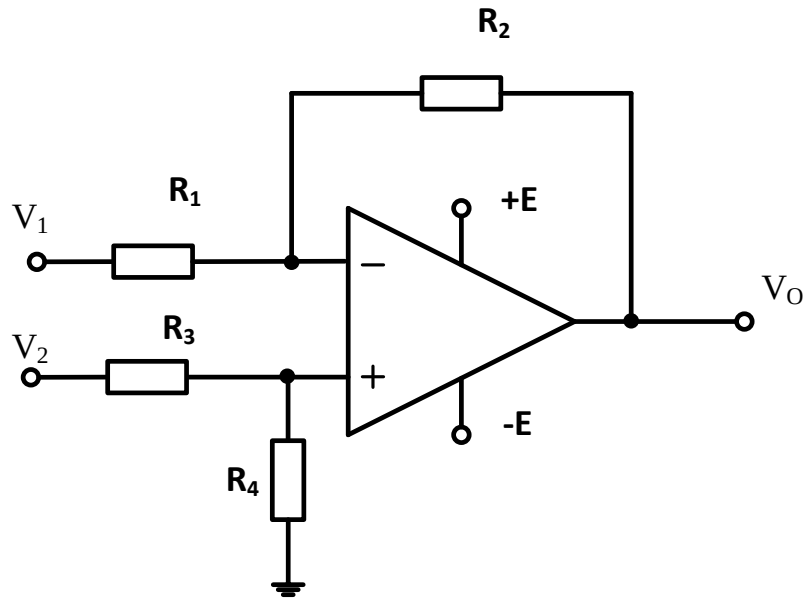


Bazele Circuitelor Electronice

Aplicații cu Amplificatoare Operaționale



Amplificator diferențial



$$R_1 = R_2 \Rightarrow v^+ = v^-$$

divisor de
tensiune

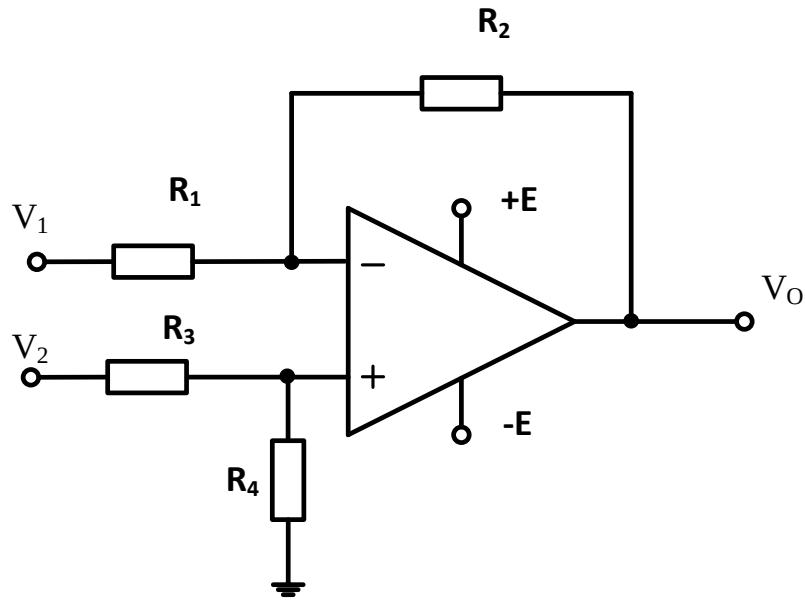
T. Millman

$$v^+ = \frac{R_4}{R_3 + R_4} v_2$$

$$v^- = \frac{\frac{v_1}{R_1} + \frac{v_O}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}}$$



Amplificator diferențial



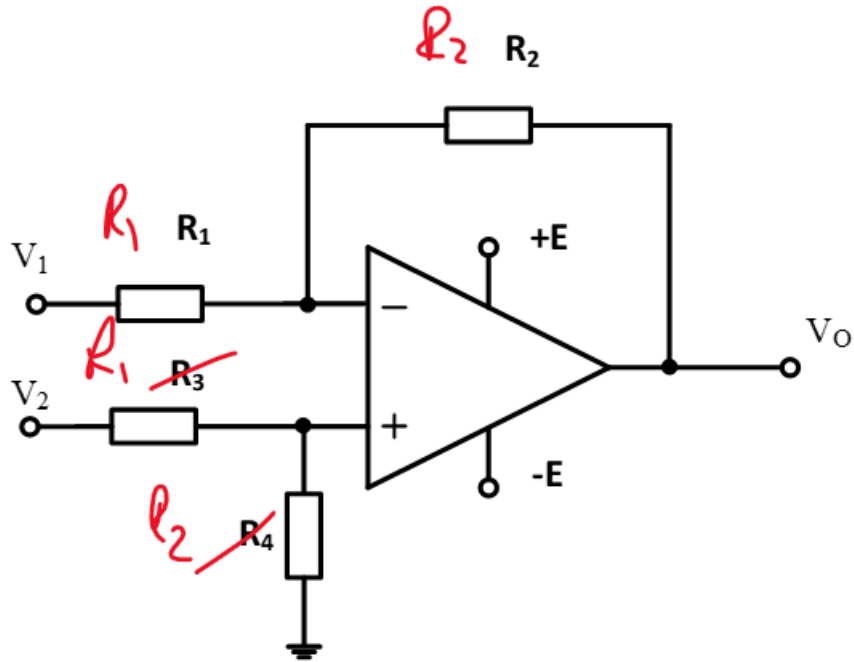
$$\Rightarrow \frac{R_4}{R_3 + R_4} V_2 = \frac{\frac{R_2 V_1 + R_1 V_0}{R_1 \cdot R_2}}{\frac{R_1 + R_2}{R_1 \cdot R_2}}$$

$$\frac{R_4}{R_3 + R_4} V_0 = \frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_0 \quad \bigg| \quad \frac{R_1 + R_2}{R_1}$$

$$\frac{R_4}{R_1} \frac{R_1 + R_2}{R_3 + R_4} V_0 = \frac{R_2}{R_1} V_1 + V_0$$

$$\Rightarrow V_0 = \frac{R_4}{R_1} \frac{R_1 + R_2}{R_3 + R_4} V_2 - \frac{R_2}{R_1} V_1$$

Amplificator diferențial

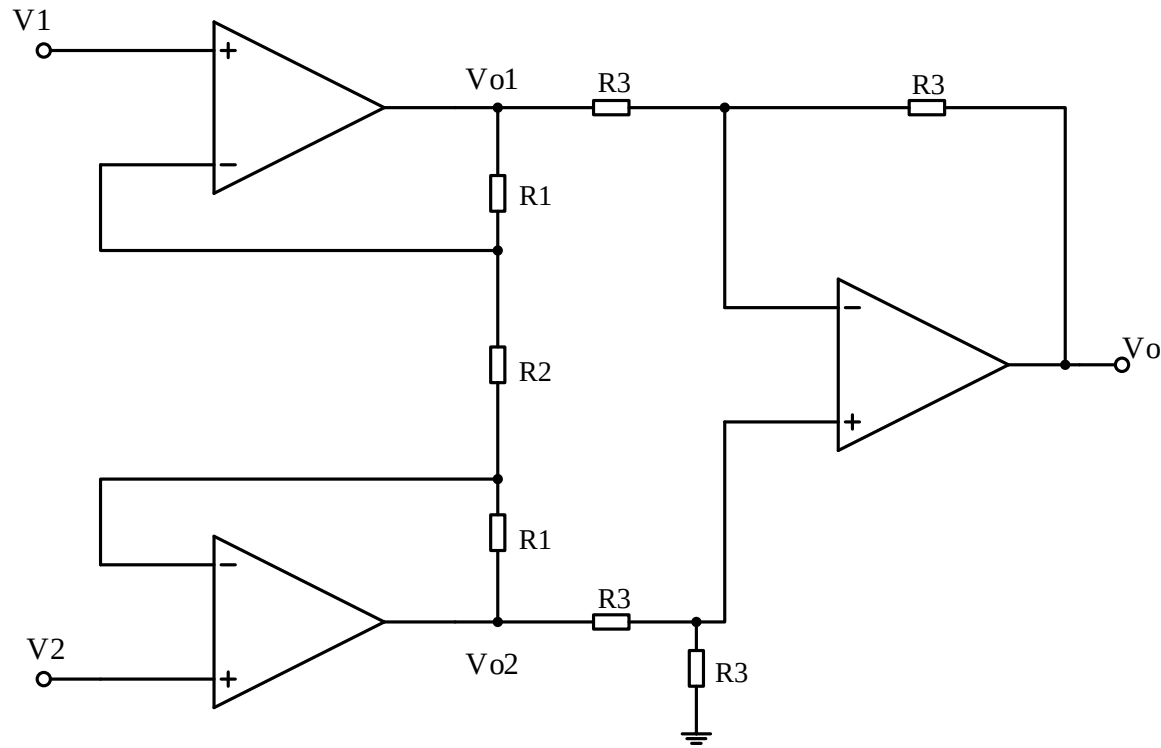


pt $R_1 = R_3$ și $R_2 = R_4$

$$V_o = -\frac{R_2}{R_1} (V_1 - V_2)$$



Amplificator de instrumentație

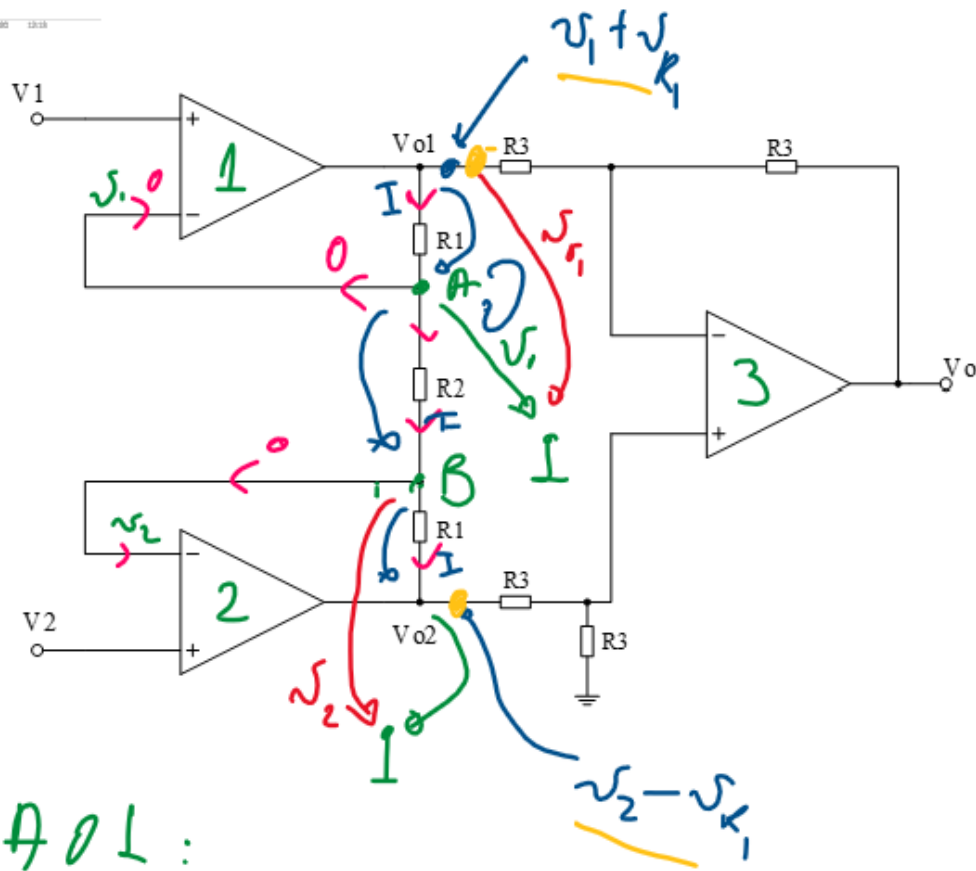


$$V_o = f(V_1; V_2) = ?$$



Amplificator de instrumentație

$$v_o = f(v_1; v_2) = ?$$



AOL:

$$\left. \begin{array}{l} v^+ = v^- \\ v^+ = v_1 \end{array} \right\} \Rightarrow v_A = v_1$$

$$v_{R1} = R_1 \cdot I$$

$$I = \frac{v_1 - v_2}{R_2} = \frac{v_1 - v_2}{R_2}$$

$$(1) \rightarrow v_{o1} = v_1 + \frac{v_1 - v_2}{R_2} R_1$$

$$(2) \rightarrow v_{o2} = v_2 - \frac{v_1 - v_2}{R_2} R_1$$

$$\Rightarrow v_o = v_1 + \frac{v_1 - v_2}{R_2} R_1$$

$$v_{o2} = v_2 - \frac{v_1 - v_2}{R_2} R_1$$

AOL:

$$\left. \begin{array}{l} v^+ = v^- \\ v^+ = v_2 \end{array} \right\} \Rightarrow v_B = v_2$$

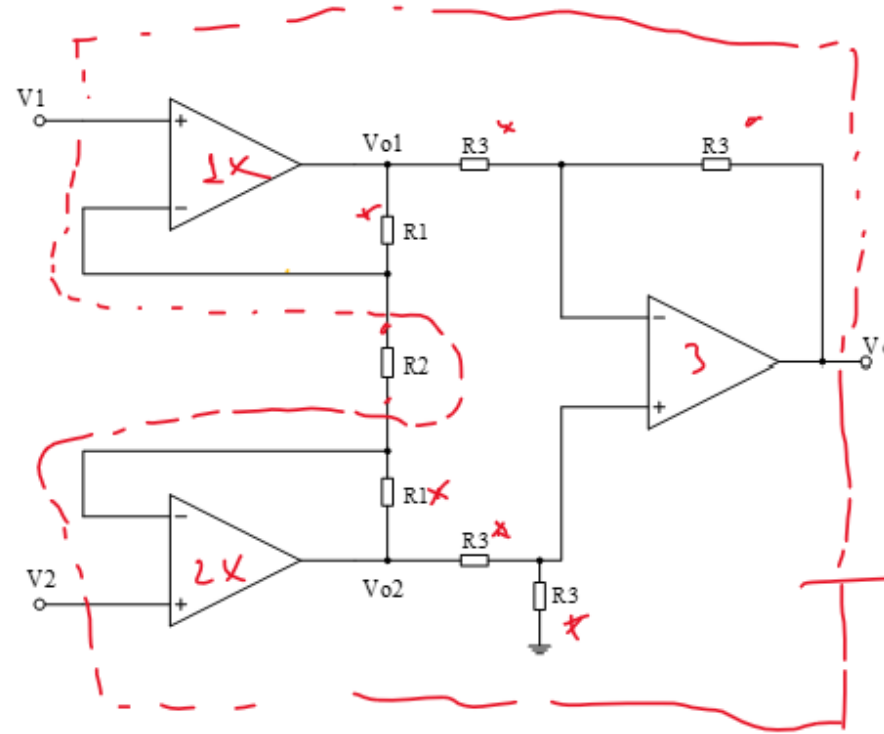
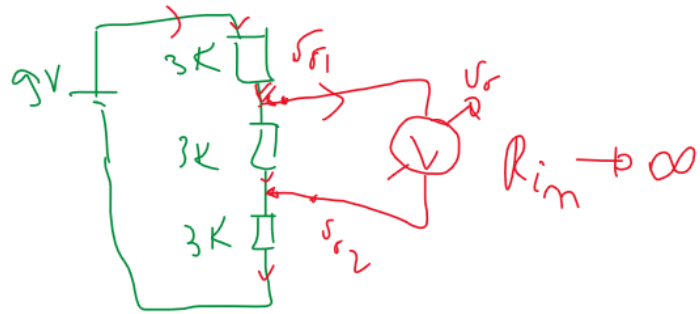
$$v_2 = v_{R1} + v_{o2}$$

$$\Rightarrow v_{o2} = v_2 - v_{R1} \quad (2)$$



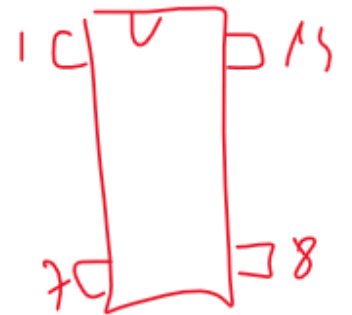
Amplificator de instrumentație

$$v_o = f(v_1; v_2) = ?$$



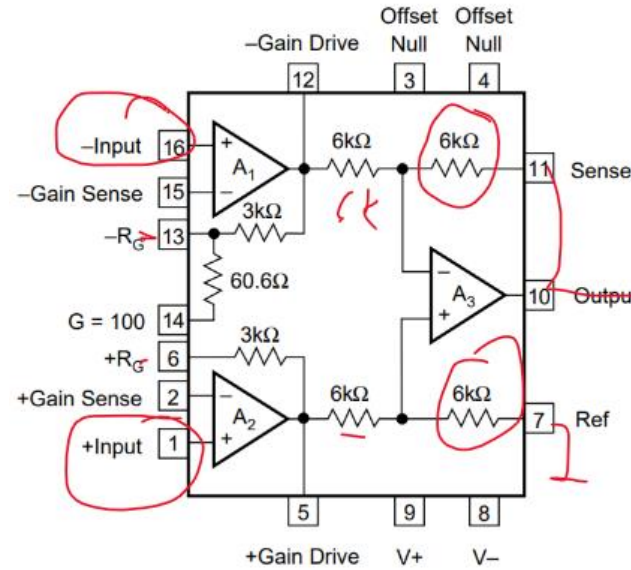
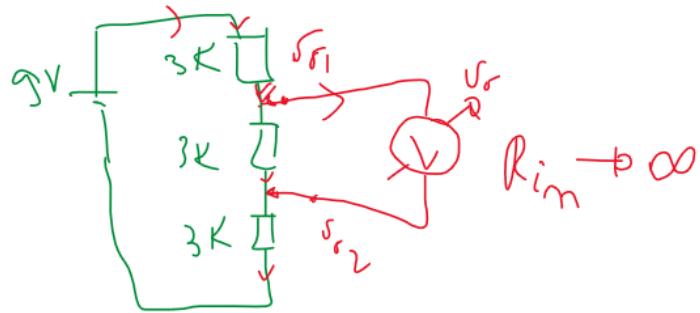
$$v_o = - \left(1 + 2 \frac{R_1}{R_2} \right) (v_1 - v_2)$$

TL084



Amplificator de instrumentație

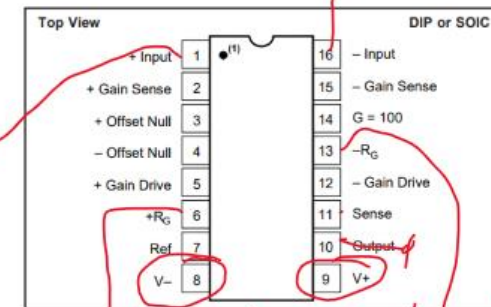
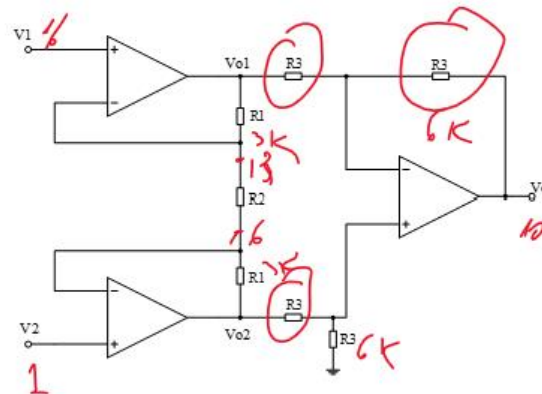
$$v_o = f(v_1; v_2) = ?$$



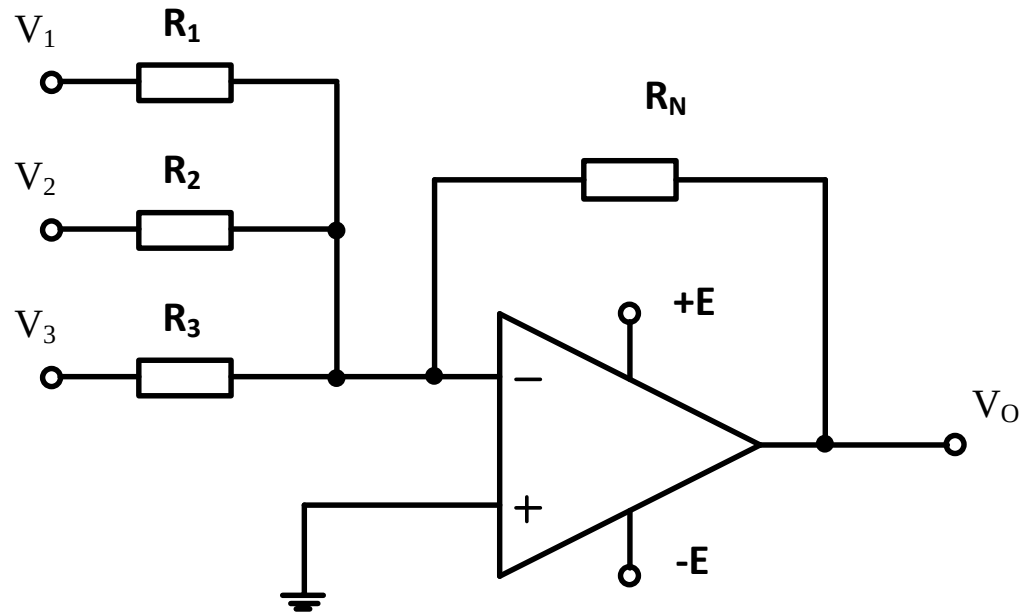
INA103

$$v_o = - \left(1 + 2 \times \frac{3k \cdot 2}{R_G} \right) (v_1 - v_2)$$

$$R_2 = R_G$$



Sumator

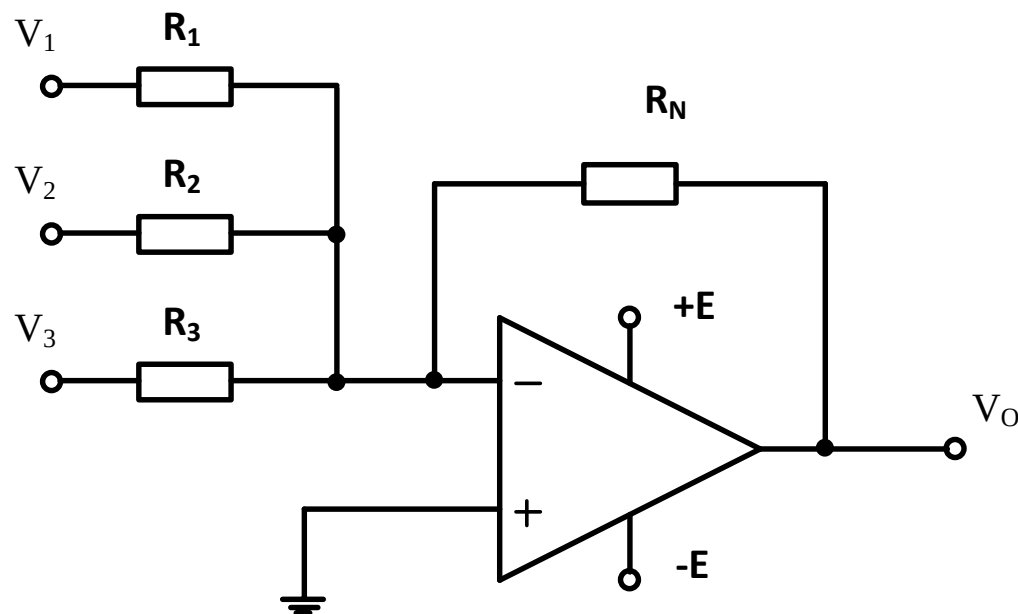


$$V_O = f(V_1, R_2, R_3) = ?$$



Sumator

$$V_o = f(v_1, v_2, v_3) = ?$$



$$R_N \rightarrow v^+ = v^-$$

$$v^+ = 0$$

$$v^- = \frac{\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} + \frac{v_o}{R_N}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_N}}$$

$$\Rightarrow \frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} + \frac{v_o}{R_N} = 0$$

$$v_o = -R_N \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} \right)$$

$$v_o = -(\alpha v_1 + \beta v_2 + \gamma v_3), \quad \alpha = \frac{R_N}{R_1}$$

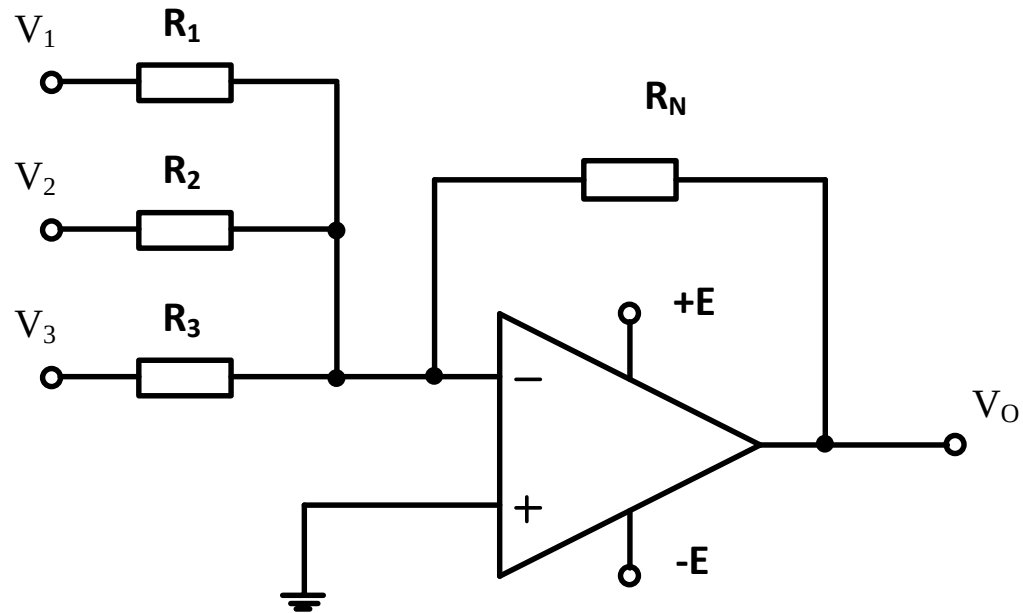
$$\beta = \frac{R_N}{R_2}$$

$$\gamma = \frac{R_N}{R_3}$$



Sumator

$$v_o = f(v_1, v_2, v_3) = ?$$



$$1) R_1 = R_2 = R_3 = R \neq R_N$$

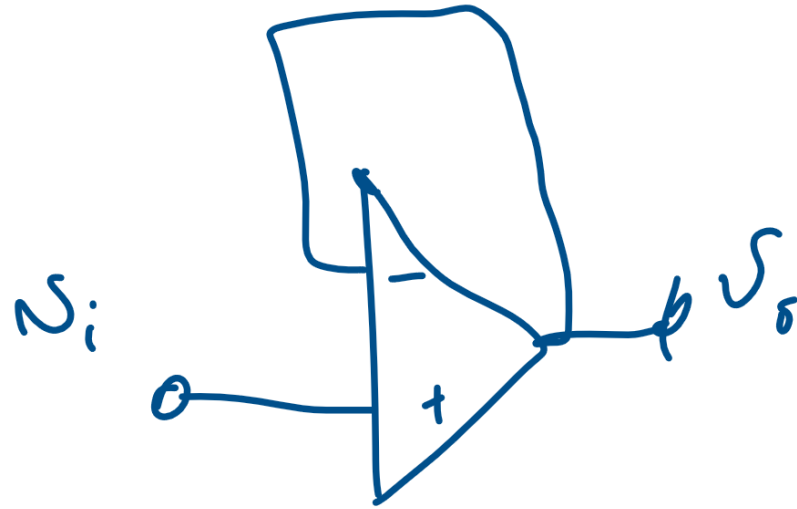
$$v_o = -\frac{R_N}{R} (v_1 + v_2 + v_3)$$

$$\therefore R_1 = R_2 = R_3 = R_N$$

$$-E < v_o = - (v_1 + v_2 + v_3) < E$$



Repetor

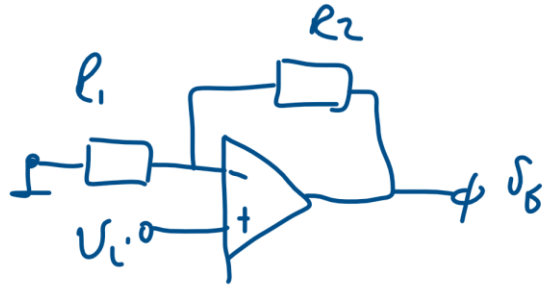


$$A = 1$$

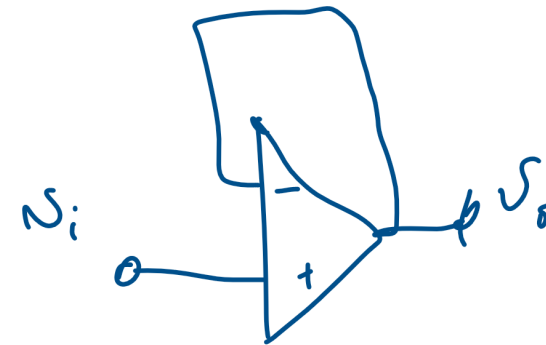
$$v_o = v_i$$



Repetor

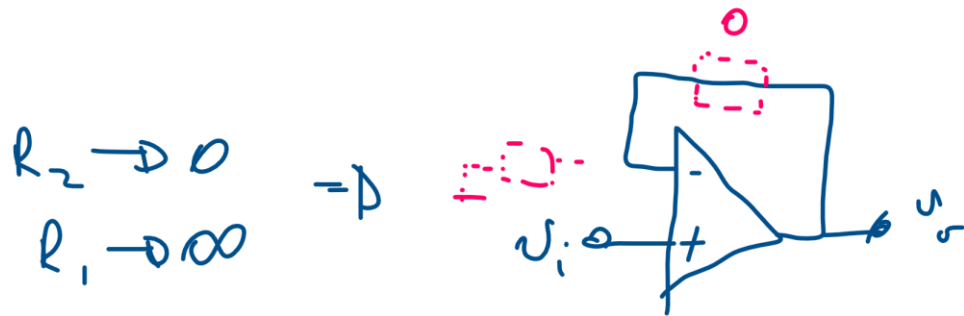


$$A = \frac{v_o}{v_i} = 1 + \frac{R_2}{R_1} = 1 + \frac{R_2}{R_1}$$

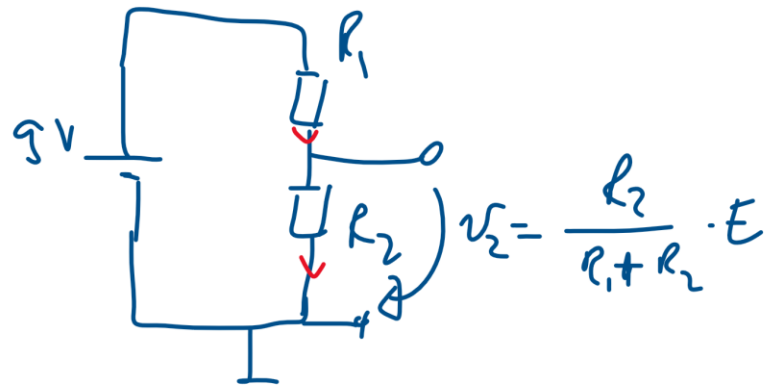
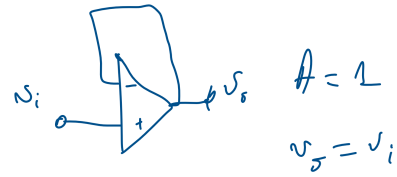


$$A = 1$$

$$v_o = v_i$$



Repetor



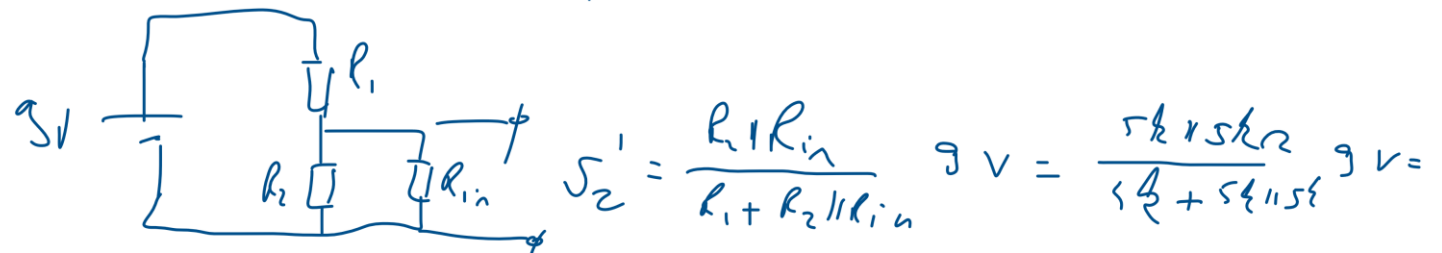
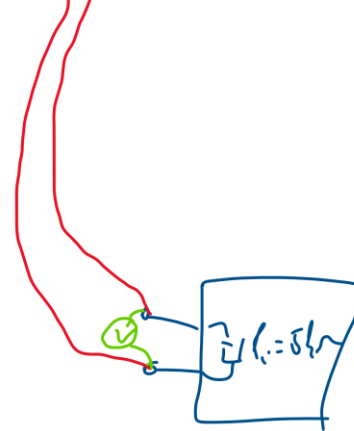
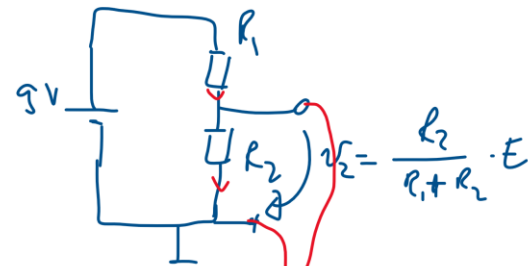
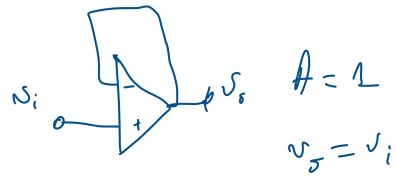
$$R_2 = 5k\Omega$$

$$R_1 = 4k\Omega$$

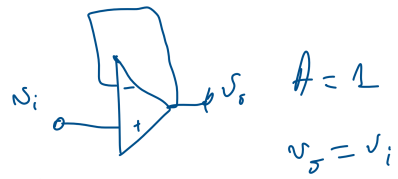
$$\Rightarrow v_2 = \frac{5k\Omega}{5k\Omega + 4k\Omega} \cdot 9V =$$

$$\frac{5V}{1}$$

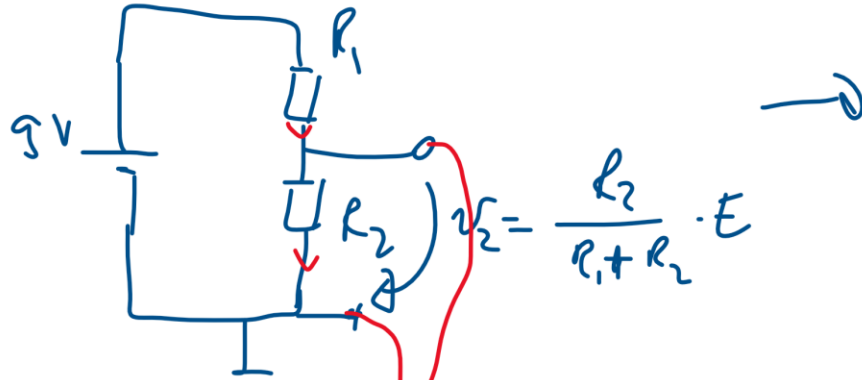
Repetor



$$v_2' = \frac{2.5k}{6.5k} \cdot 9V = 3.4V$$



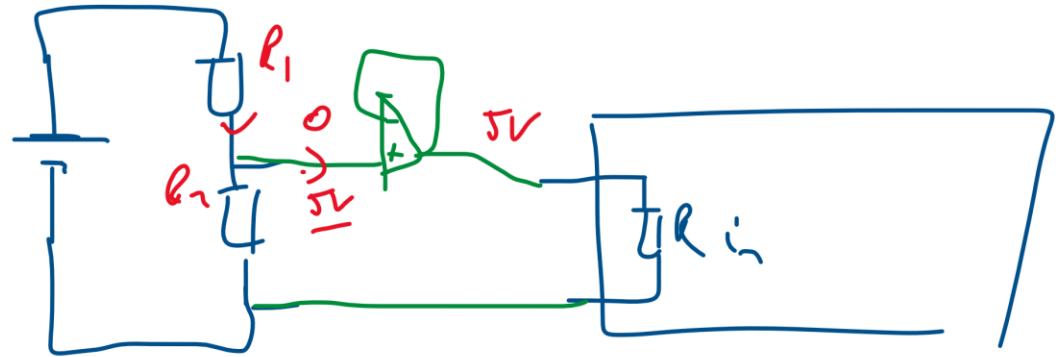
Repetor



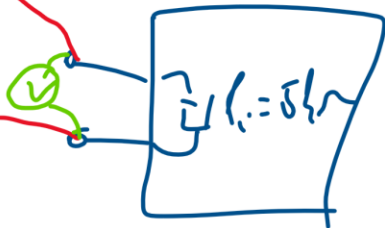
$$R_2 = 5k\Omega$$

$$R_1 = 4k\Omega$$

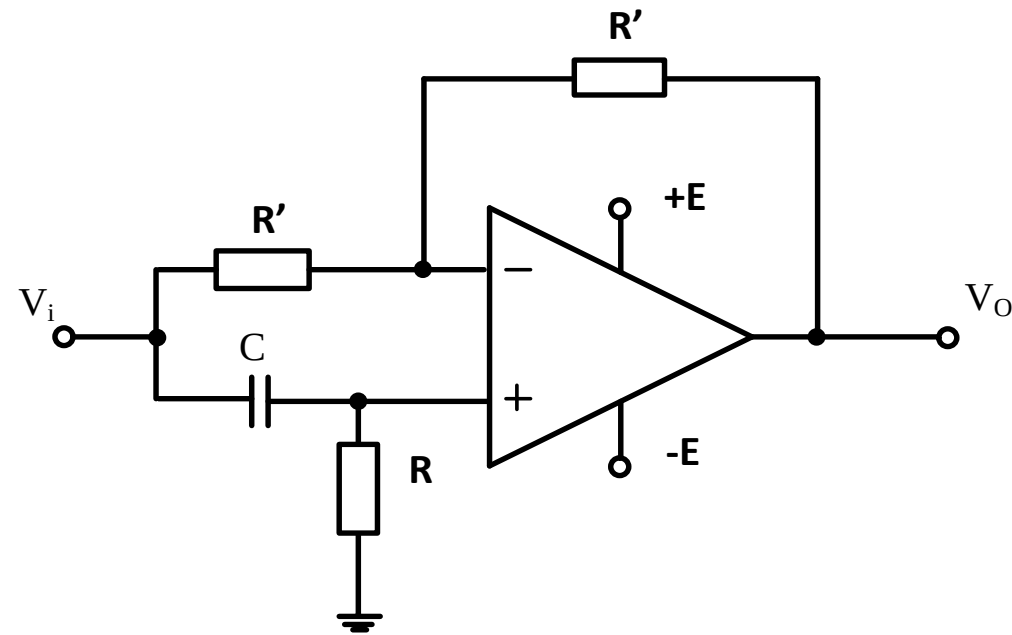
$$\Rightarrow v_2 = \frac{5k\Omega}{5k\Omega + 4k\Omega} \cdot 9V =$$



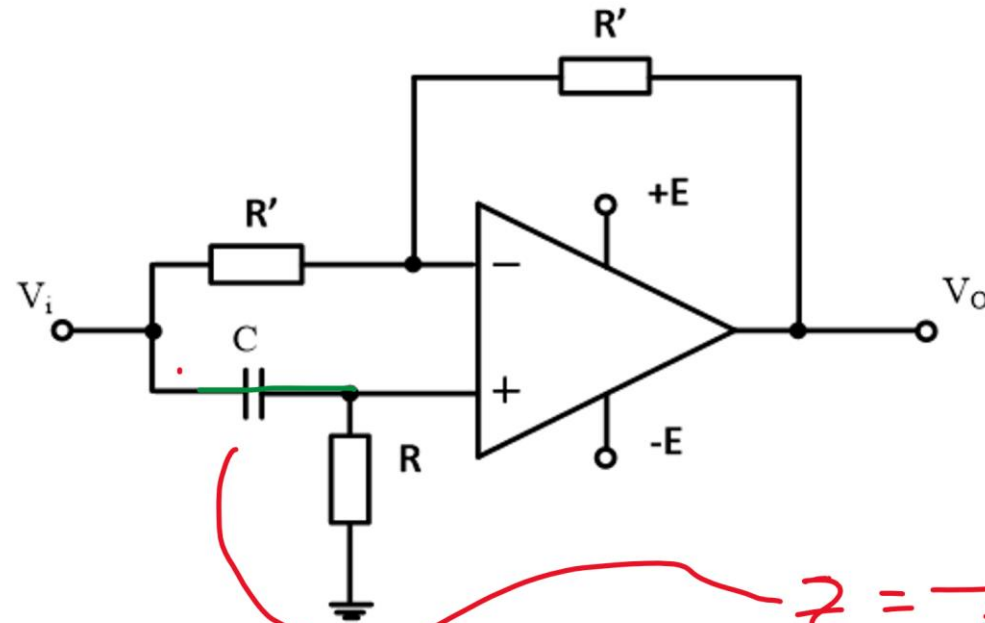
$$5V$$



Filtru Trece Tot



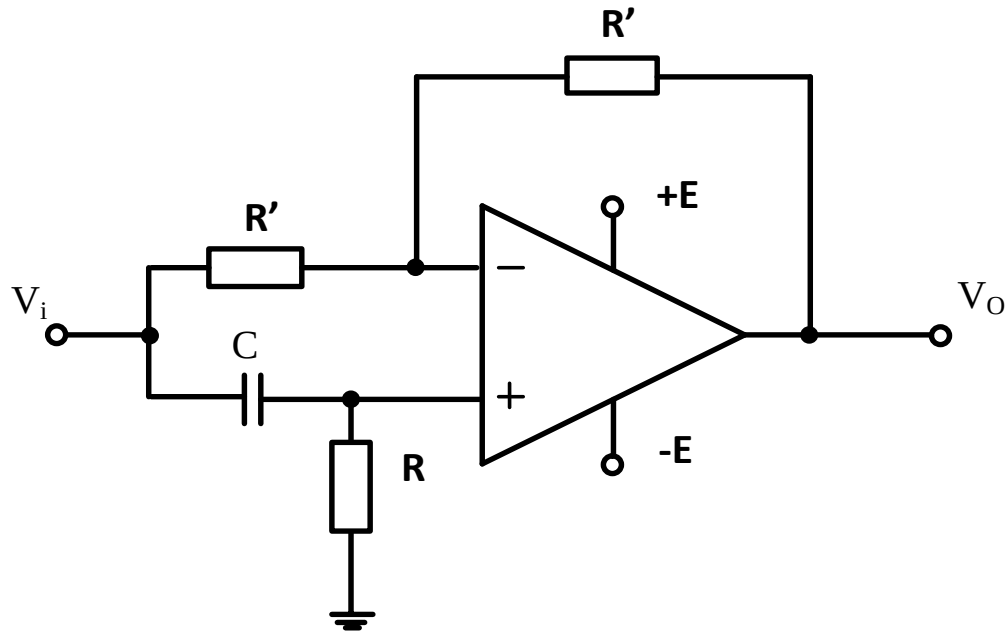
Filtru Trece Tot



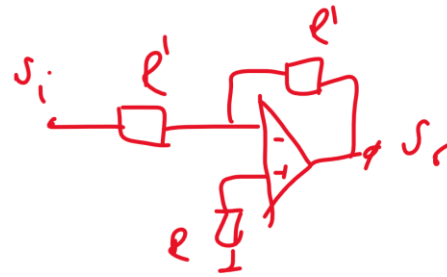
$$Z = \frac{1}{j\omega C} = \frac{1}{j2\pi f C}$$



Filtru Trece Tot



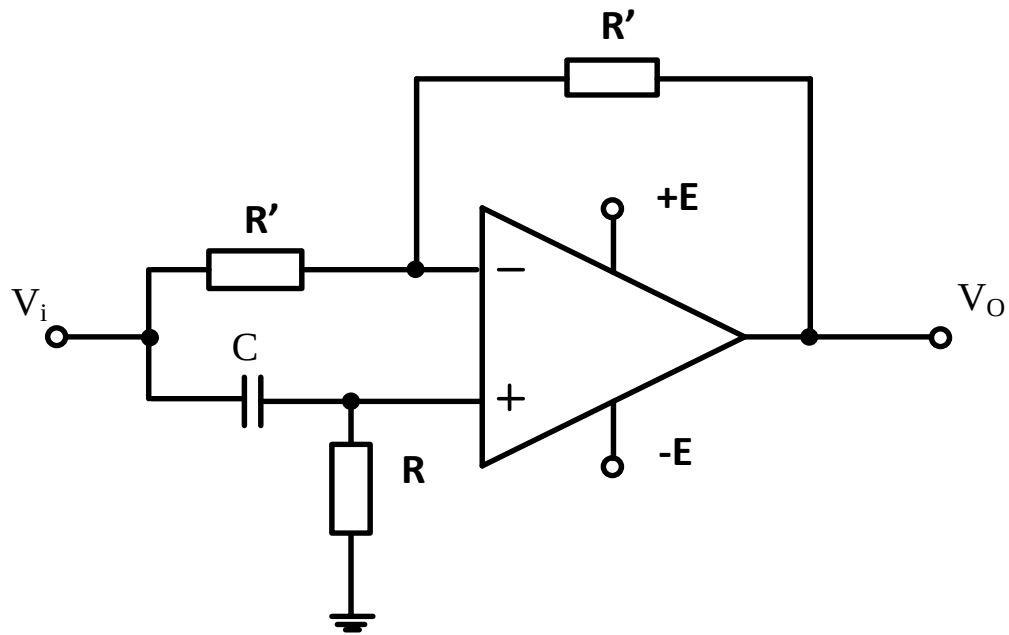
Pr. $f_i \rightarrow 0$
 $\rightarrow Z_C \rightarrow \infty$



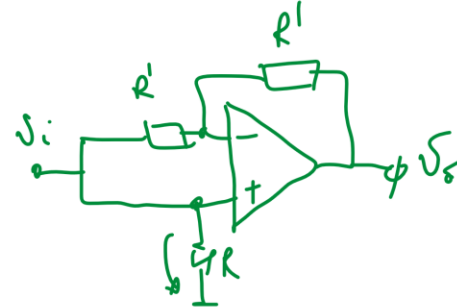
$$A = - \frac{R_l}{R_1} = - \frac{R'}{R} = -1$$



Filtru Trece Tot



pp $f_i \rightarrow \infty \Rightarrow Z_c \rightarrow 0$



$R \Rightarrow V^+ = V^-$

$$V^+ = V_i$$

$$V^- = \frac{\frac{V_i}{R'} + \frac{V_o}{R'}}{\frac{1}{R'} + \frac{1}{R'}}$$

$$\underline{V_i} = \frac{\frac{V_i + V_o}{R}}{\frac{2}{R}} = \frac{V_i + V_o}{2} \quad \text{r2}$$

$$2 V_i = V_i + V_o \Rightarrow V_o = V_i$$

$$A = \frac{V_o}{V_i}$$

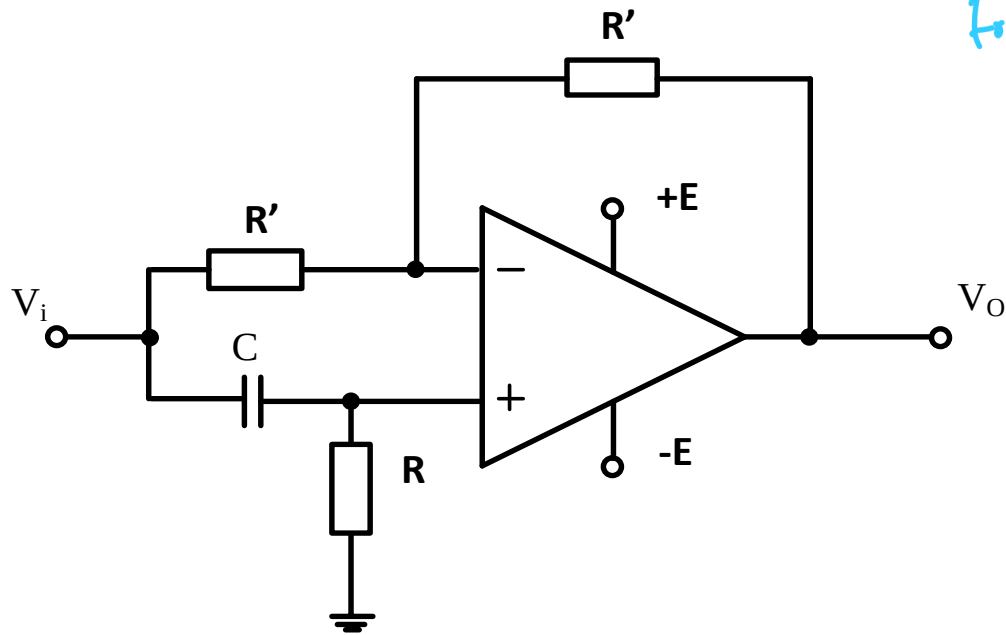
$$\boxed{A = +1}$$

Filtru Trece Tot

$$R = 10k$$

$$C = 100n$$

$$f_c = \frac{1}{2\pi RC} = \frac{0,159}{1000\mu} = 159Hz$$



$$f \rightarrow 0 \rightarrow A = -1$$

$$f \rightarrow \infty \rightarrow A = +1$$

$$|A| = 1, \forall f$$

